

Ghost in the VR Shell: Capturing Spectral Cardio-Respiratory Rates from Subtle VR Device Movements

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Abstract

This work examines the estimation of heart rate and respiratory rate using only the kinematic data captured by a consumer-grade standalone VR devices. The high-resolution motion tracking offered by these devices creates an opportunity for indirect vital sign detection through spectral analysis of subtle VR device movement data. In our study, kinematic data were collected from a Meta Quest 3 head-mounted display, controllers and MX Ink pen across multiple posture configurations (e.g., seated, standing, lying down), both at rest and after moderate exercise. These postures emulate real-world XR scenarios for rest, fitness, and meditation. The collected data was processed using what we refer to as the *Ghost* approach, a simple yet effective method that applies a Fast Fourier Transform to capture the spectral components associated with respiratory and cardiac rhythms. Ground-truth biosignals were simultaneously recorded using wearable physiological sensors for validation. Results clearly reveal that both heart rate and respiratory rate can be reliably estimated from subtle micro-movements in the head-mounted display, VR controllers, or VR pen, revealing the potential for non-contact physiological monitoring within immersive environments. Finally, we demonstrate a use case of a VR stethoscope, where a standard VR controller is repurposed to estimate heart and respiratory rates.

CCS Concepts

• **Computing methodologies** → **Motion processing**; • **Human-centered computing** → **Virtual reality**; **Ubiquitous and mobile computing**.

Keywords

Standalone VR devices, VR pen, Kinematic data, Biosignal processing, Heart rate, Respiratory rate

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1 Introduction

Nowadays, Virtual and Augmented Reality (VR/AR) devices combine multiple embedded sensors and precise motion tracking with biosignal processing to create enriched immersive experiences [33] that are transforming fields in healthcare [24], mental health [3], wellness [34], fitness [23], sports performance [15, 22], emotion recognition [31], pulmonary rehabilitation [9, 19], phobia treatment [30], and entertainment [21]. For example, in therapeutic settings, real-time stress monitoring can support personalized interventions for anxiety and post-traumatic stress disorder by adapting stimuli to guide users toward relaxation [11, 27]. In gaming and entertainment, physiological feedback can improve immersion by dynamically adjusting gameplay difficulty based on the user's stress or excitement levels [21]. Additionally, adapting VR content in response to biosignals can help reduce cybersickness [26]. The effectiveness of these



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experiences relies heavily on user comfort, emotional engagement, and a strong sense of presence, all of which can be assessed through biosignals and derived physiological parameters such as heart rate (HR) and respiratory rate (RR), both of which serve as proxies for emotional and cognitive states in VR [13, 25].

In general, VR devices lack clinically rated sensors or dedicated hardware for biosignal acquisition and processing. Integrating such capabilities is challenging, as traditional physiological signals (e.g., electrocardiogram, plethysmogram) typically require specialized hardware, complex circuitry, and careful calibration procedures. In addition, wearable devices usually restrict access to raw physiological data, limiting custom integration with VR platforms and posing challenges for real-time applications. To address this challenge, recent studies have investigated leveraging sensors already embedded in VR hardware to estimate HR and RR [8, 10, 16, 17, 36]. These integrated sensing technologies offer a more seamless and cost-effective approach by utilizing existing VR sensor components such as IMUs, gyroscopes, and magnetometers. However, real-world validation of these methods remains limited [8], and most commercial VR systems continue to focus primarily on visual, auditory, and motion inputs, often leaving out the integration of biosignals [12].

This work addresses a critical gap by exploring an overlooked biosignal source in VR systems: high-resolution motion tracking data that captures the degrees of freedom (DoFs) of each device (i.e., positions and orientations), along with their derived kinematic features (i.e., velocities and accelerations).

By focusing exclusively on the subtle movements of VR devices, we aim to extract HR and RR without requiring additional sensor-related hardware. Our approach uses standard frequency-domain analysis, specifically Fast Fourier Transforms, to reveal spectral components associated with physiological rhythms. We call this method *Ghost* because it captures subtle spectral traces, like hidden “ghost signals”, that reside within the natural movements of the VR devices. Our *Ghost* approach not only simplifies the sensor setup but also has the potential to enhance user experience in popular VR scenarios (e.g., rest periods during VR fitness or when adapting exercise to target intensity, restful health tracking, guided meditation) ultimately advancing immersive VR interactions through increased embodiment and presence. We present results for the Meta Quest 3 head-mounted display (HMD), its controllers, and also the MX Ink stylus [14]. We further introduce a VR stethoscope use case, showcasing how a standard VR controller can be repurposed to estimate HR and RR through chest placement.

2 Related Work

Recent advances in micro-electromechanical systems (MEMS), inertial sensors, and computer vision have expanded the capabilities of VR headsets and wearable devices for non-invasive physiological monitoring. Studies have demonstrated the feasibility of tracking biosignals and extracting vital parameters such as HR and RR using VR headset inertial sensors, ballistocardiography (BCG), and contactless optical techniques such as remote photoplethysmography (rPPG). Below, we categorize prior work based on sensing modality and system type.

Early explorations in inertial sensing for physiological estimation using off VR headsets were conducted by Floris et al. [10]. This

study used the Oculus Go to estimate HR and RR in 30 healthy participants across three postures: sitting, standing, and supine. Data was collected via embedded accelerometer and gyroscope sensors, along with simultaneous 1-lead ECG as ground truth. Spectral analysis techniques revealed that gyroscope data outperformed accelerometer inputs, with HR estimation most accurate in the sitting position ($R^2 = 0.91$) and RR estimation best in the supine posture (98% feasibility). This validated posture-sensitive physiological signal extraction via MEMS sensors embedded in VR hardware but did not explore VR DoFs or kinematic features whatsoever.

Building on this, Solbiati et al. [28] introduced rotational and linear kinetic energy metrics derived from Oculus Quest gyroscope and accelerometer data. They estimated HR from 1-minute recordings across supine, sitting, and standing postures in eight participants. Rotational kinetic energy, especially from the gyroscope, yielded accurate HR estimates comparable to ECG references in the sitting and supine postures. However, performance declined in the standing posture due to involuntary head movements.

Further refining the inertial sensing approach, Solbiati et al. [29] compared embedded VR gyroscopes (VRG) with high-performance gyroscopes (HPG) in a cohort of 9 healthy participants. HR was derived from head-BCG signals while lying down. Pitch and roll axes produced higher-quality signals than yaw, and the HPG outperformed VRG in beat detection. These results underscore the benefits of high-grade inertial sensors for accurate physiological monitoring in VR. Complementing these works, Carpi et al. [6] proposed a web-based, device-independent VR respiratory training system using head-BCG data to estimate RR, tested by 26 participants that followed guided breathing exercises while RR was estimated in quasi-real time using a grey-box model optimized against ECG-derived RR. The algorithm achieved <4% estimation error and was robust to motion noise.

In contrast to inertial approaches, optical sensing methods focus on contactless analysis of physiological signals using video. Pagano et al. [18] utilized HMD-mounted cameras to estimate HR from facial regions. Using enhanced video magnification and a Meta-rPPG technique combined with support vector machines, they achieved superior beats-per-minute prediction compared to baseline models. This approach requires front-facing cameras on HMDs and does not rely on inertial data.

Together, these studies illustrate the growing potential of both inertial and optical sensing technologies for non-contact physiological monitoring. Inertial methods, especially gyroscope-based BCG in VR and wearable form factors, demonstrate robust performance for HR and RR estimation. Optical approaches, such as rPPG and motion magnification, offer complementary capabilities with broader application in contactless and camera-based systems. However, we identified two key gaps in the existing literature: (i) no prior study has examined the use of high-resolution motion tracking data acquired from VR devices (i.e., the DoFs and derived kinematic features) as a potential source of physiological parameter extraction; and (ii) the integration of consumer-grade VR peripherals, such as VR pens, for physiological inference remains unexplored.

Finally, it's worth noting that several commercial VR apps incorporate breath-driven interaction using motion signals, yet most fall short in reliability and user experience. A kinematics-based solution to extract HR and RR, as the one we explore in our *Ghost*

approach, may provide the natural, accurate interaction these apps currently lack. Titles such as *Zengence*¹, *BreathTech*², and *Realms of Flow*³ appear to rely on audio cues, likely using microphone input to detect inhalations and exhalations, but offer limited accuracy and robustness. *FlowBourne*⁴ attempts to use the motion of a controller placed on the abdomen to detect abdominal breathing. However, interaction is coarse, requires exaggerated respiratory effort causing user discomfort. Similarly, *Realms of Flow* combines vertical head tilts, controller movement, and sound, yet suffers from the same limitations. *Novobeing*⁵, while providing guided breathing experiences, does not extract actual breathing signals from device kinematics. In summary, these apps lack reliable, real-time respiratory detection, requiring deep breathing or exaggerated motion, and suffer from unintuitive mappings between user input and in-game responses.

3 Methods

3.1 Evidence of Micro-Movements

This study began with a simple yet revealing observation: even in the absence of intentional movement, VR devices display subtle micro-movements that are perceptible to the naked eye. For instance, when a user steadily holds a controller or a VR pen and points to a flat panel, then subtle, periodic fluctuations of the ray-casted cursor become visible. So, in the absence of large ranges of motion, the cursor appears slightly unstable, gently drifting up and down or side to side. These minute oscillations reflect the body's involuntary physiological processes, suggesting that high-resolution motion tracking in VR systems is sensitive enough to capture micro-movements linked to heartbeats and/or respiration.

3.2 Kinematic Features and Vital Parameters

Table 1 showcases the collected kinematic data that include position (X, Y, Z) and orientation (pitch, roll, yaw) measurements, along with linear velocity, linear acceleration, angular velocity, and angular acceleration, which were captured from each VR device (i.e., HMD, left and right controllers, and VR pen). This results in a total of $4 \times 6 \times 3 = 72$ signals (number of devices $\times$ kinematic features $\times$ coordinates) per VR activity. As for the physiological parameters of interest, we considered HR and RR reported in beats per minute (bpm) and cycles per minute (cpm), respectively.

3.3 Subjects

A total of 15 healthy adult subjects (mean age 23.0 ± 6.3 years; 8 males and 7 females) participated in this study. Eligibility criteria included being over 18 years old, non-smokers, and free from any musculoskeletal, cardiac, or respiratory conditions, as well as any other health issues that could interfere with safe completion of the experimental activities. The study protocol was approved by the local Ethics Committee (REF 20/2025 (CE-IST)) and carried out in accordance with the Declaration of Helsinki.

Table 1: List of kinematic features and vital parameters analyzed in the *Ghost* study.

Kinematic Feature	Unit
Position (X, Y, Z)	m
Orientation (Pitch, Roll, Yaw) - Euler angles	deg
Linear Velocity (X, Y, Z)	m/s
Angular Velocity (Pitch, Roll, Yaw)	rad/s
Linear Acceleration (X, Y, Z)	m/s ²
Angular Acceleration (Pitch, Roll, Yaw)	rad/s ²
Vital Parameter	Unit
Heart Rate	bpm
Respiratory Rate	cpm

3.4 Apparatus

We considered the following VR devices: a Meta Quest 3 HMD (Figure 1 - A), two Meta Quest 3 Touch Plus controllers (Figure 1 - B), and a Logitech MX Ink Mixed-Reality stylus (Figure 1 - C). The Quest 3 HMD integrates a 6-DoF IMU (3-axis gyroscope and accelerometer), four 400×400 px infrared tracking cameras, two 4 MPixels RGB passthrough cameras, a patterned-IR depth sensor, and a proximity sensor, thus, providing high-frequency head orientation, along with inside-out positional tracking, mixed-reality video, and depth mapping. Each Touch Plus controller houses a 6-DoF IMU, a Hall-effect magnetometer, capacitive touch sensors on buttons and thumbsticks, and infrared tracking markers, thus enabling precise hand pose estimation, low-drift heading stabilization, finger-touch/grip detection, and full 6-DoF controller tracking. The MX Ink stylus incorporates a 6-DoF IMU, a high-resolution pressure sensor (4096 levels), a $\pm 60^\circ$ tilt-detection mechanism, and a capacitive grip sensor, together with an onboard haptic actuator, hence delivering precise tip motion and orientation, continuous pressure and tilt data, grip state, and tactile feedback.

Although these VR devices are equipped with several sensors, this study focused exclusively on collecting and analyzing the kinematic data listed in Table 1. Kinematic data were obtained via the OpenXR (v1.1.37) and LibOVR (v1.26) APIs, interfaced through custom VR apps developed in Godot (v4.4). For biosignal acquisition, we resorted to a ScientISST Core3 board [20] (Figure 1 - D). The Core3 features six 12 bit and two 24 bit analog inputs, two 1 bit digital inputs, two 1 bit digital outputs, one 8 bit output, and I²C/SPI interfaces. It supports sampling rates up to 16 kHz and communicates via Bluetooth, Wi-Fi, or USB-serial. Electrocardiographic signals were recorded using an ECG sensor (Figure 1 - E), which provides a fixed gain of 1100, operates within a 0.5 Hz – 40 Hz bandwidth, has an input impedance of 100 G Ω , and achieves a common-mode rejection ratio of 110 dB. Thoracic respiratory activity was measured using a piezoelectric transducer embedded in an elastic chest band (Figure 1 - F), which detects Hall-effect voltage changes induced by chest expansion and contraction, serving as a surrogate for respiratory cycles [5].

3.5 Activities

Prior to data acquisition, all subjects provided written informed consent following a detailed briefing on the experimental protocol.

¹<https://www.meta.com/en-gb/experiences/zengence-take-aim-with-every-breath/7982150971811780/>

²<https://breathtechapp.com/>

³<https://www.meta.com/experiences/realms-of-flow/6062798767111272/>

⁴<https://flowborne.com/>

⁵<https://www.novobeing.com/>

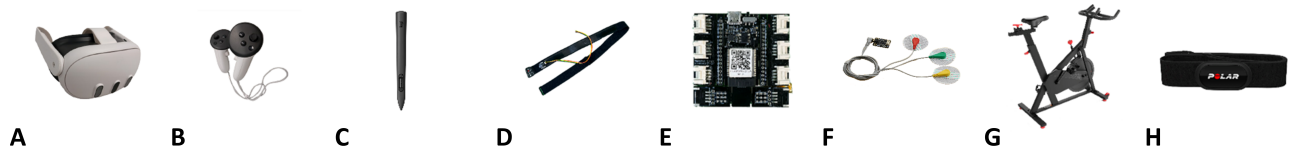


Figure 1: Apparatus used in our study for data acquisition: (A) Meta Quest 3 Head Mounted Display, (B) Meta Quest 3 Touch Plus controllers, (C) Logitech MX Ink Mixed-Reality, (D) Piezoelectric sensor for RR measurement, (E) ScientISST Core3 board, (F) ECG sensor, (G) Stationary Bicycle, and (H) Polar HR monitor.

They were then instrumented with a 3-lead ECG configuration using gel electrodes positioned on the right clavicle (R), left clavicle (L), and lower right chest (G), along with a respiratory band secured beneath the sternum (Figure 2). Once instrumentation was complete, participants were equipped with the VR headset and controllers or pen and completed four sets activities, each set combining various postures (Standing, Sitting, Lying, Drawing Standing, and Drawing Seated) with distinct breathing cadences (Self-Paced, Guided Breathing at 6 cycles per minute (cpm), Guided Breathing at 12 cpm, and Post-Exercise), as illustrated in Figure 3. This structured sequence allowed us to assess cardiovascular and respiratory dynamics during self-paced breathing, two guided breathing rates, and recovery following moderate exercise. Note that all activities emulate real-world XR scenarios for rest, fitness, and meditation.

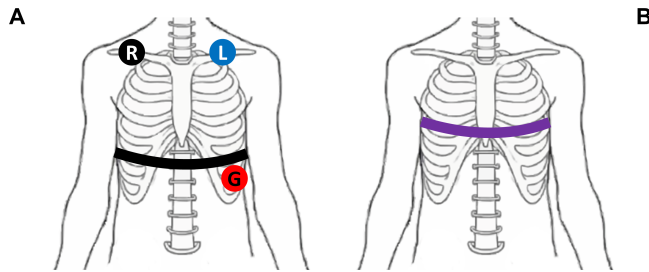


Figure 2: Illustration showing the (A) placement of ECG gel electrodes and the respiratory rate measurement band secured just below the sternum on a study participant, and the (B) placement of the Polar heart rate monitor used during the stationary bicycle exercise.

In the first set of activities, we explored self-paced respiration, where participants assumed four postures (standing, seated, supine, and grasping the MX Ink stylus, both resting on a table and in a drawing posture) while we recorded baseline data. In the Standing posture (first), participants stood upright with their arms at their sides. In the Sitting posture (second), they sat with their arms resting on their thighs. For the Lying posture (third), participants lay in a supine position with arms at their sides, looking upward, either on a yoga mat or a table. In the Drawing while Standing posture (forth), participants stood while holding the pen with a flexed arm, positioning their hand near head level. Finally, in the Drawing while Seated posture (fifth), they sat at a table and held the pen.

For each posture, researchers first completed instrumentation and then displayed the virtual environment through the HMD. Before data acquisition, participants were instructed to breathe at

a self-paced respiratory rate. After an audio cue, data collection began and lasted for 2 minutes. No breathing guidance was provided during this activity. A 2-minute rest period followed each posture. The same sequence and rest intervals were applied across all tasks.

In the second set of activities, participants engaged in guided respiration by following a visual cue within the virtual environment, which consisted of a ring that expanded and contracted to prompt breathing at a controlled pace (6 cpm). This set of activities was performed across the same four postures defined in the first activity. The third set of activities followed the same protocol as the first (self-paced) and second (guided at 6 cpm), but with participants instructed to perform guided breathing at a faster rate (12 cpm).

For the final activity, participants were temporarily unequipped from VR devices. They were then fitted with a Polar heart rate monitor⁶ and used a stationary bicycle until reaching 65% of their age-predicted maximum heart rate (calculated as $(220 - \text{age}) \times 0.65$). Following exercise, participants were re-equipped with the VR headset and controllers, then performed only the standing posture with self-paced respiration breathing for two minutes.

3.6 Signal Processing

For each device, we recorded the kinematic data provided by the APIs in the form of three-dimensional position, orientation, linear and angular velocities, and linear and angular accelerations, all at a 90 Hz sampling rate. Prior to spectral analysis, orientation was collected as quaternion data but converted to Euler angles, and all time-series signals were uniformly resampled.

Our *Ghost* approach was designed with the same guiding principles that made the 1 ϵ filter algorithm [7] widely popular in the HCI community: simplicity, ease of implementation, effectiveness, and low computational overhead for real-time signal processing. These shared characteristics make *Ghost* particularly attractive for XR apps where responsiveness is essential.

The *Ghost* spectral analysis was performed using the Fourier Transform, chosen for its reduced computational cost of $O(N \log N)$, as illustrated in Figure 4. For HR estimation, spectral components in the 0.8 Hz - 2 Hz range were analyzed, while the 0.1 Hz - 0.4 Hz band was used for RR estimation. Dominant frequency components extracted from each signal were compared against ground-truth HR and RR obtained from biosignals recorded with the ScientISST Core3 at a 100 Hz sampling rate. To enable continuous monitoring, analysis was carried out on sliding 30 s windows with a 1 s step size. This configuration introduces an initial latency of 30 s (to fill the first window) but then provides quasi real-time updates at 1 s intervals. ECG data were filtered with a 0.67 - 45 Hz bandpass filter

⁶<https://www.polar.com/en/sensors/h10-heart-rate-sensor?sku=92075957>



Figure 3: Overview of the five postures evaluated in this study. **Standing:** participant stands upright with arms relaxed at their sides; **Sitting:** participant sits on a chair with arms resting comfortably on the thighs; **Laying (Supine):** participant lies flat on their back—either on a yoga mat or a table—with arms at their sides and gaze directed upward; **Drawing while Seated:** participant remains seated at a table, holding the pen in a natural drawing posture; **Drawing while Standing:** participant stands with the arm flexed and the hand holding the pen positioned at head level.

to remove baseline drift and high-frequency noise. Heart Rate was computed from R-R intervals detected in the filtered ECG waveform and subsequently converted to Hz for comparison with inertial data. Respiratory signals were filtered with a 0.1 Hz - 0.35 Hz bandpass and analyzed via zero-crossing detection to estimate respiratory cycles. All biosignal processing was performed using the BioSPPy Python package [4].

3.7 Performance Evaluation

We assessed the accuracy of HR and RR estimates derived from VR device kinematic data by calculating the Root Mean Square Error, $RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \hat{x}_i)^2}$, and the Percentage Absolute Error, $PAE = \frac{1}{n} \sum_{i=1}^n \left(\frac{|x_i - \hat{x}_i|}{x_i} \right) \times 100$, where x_i denotes the ground-truth value, $\hat{x}_i$ represents the estimated value, and n is the total number of observations. These metrics were calculated between the estimates obtained from each VR device (head mounted display, left controller, right controller, VR pen) and the ground-truth signals recorded using the ScientISST Core, across all experimental activities. The total duration of each signal was 90 seconds. RMSE and PAE were calculated for overlapping 30 s windows, with a 1 s shift between consecutive windows (Figure 4).

4 Results

Table 2 summarizes the kinematic signals that yielded the most accurate HR and RR estimates across the various VR postures and breathing cadences.

The analysis of the results reveals that HR estimation was highly dependent on both posture and breathing cadence, with no single device or signal consistently outperforming others. In standing tasks, the Right Controller (RC) device achieved the highest accuracy during self-paced and 12 cpm breathing using linear acceleration features, while the HMD and Left Controller (LC) devices performed best at 6 cpm using angular acceleration. Post-exercise conditions introduced greater estimation challenges, though the HMD device remained the most accurate. In the sitting posture, the RC device consistently delivered the best performance at self-paced and 6 cpm

cadences, whereas the HMD device excelled at 12 cpm. For lying tasks, RC and LC devices using linear velocity features achieved the lowest errors, highlighting their robustness in low-motion settings. In drawing activities, sensor performance varied by context: the HMD device performed best in standing drawing tasks across all cadences, while the Pen device was more effective during seated drawing at lower cadences, with HMD surpassing it at 12 cpm.

In contrast, RR estimation showed more consistent patterns across conditions. Accuracy generally improved with increasing breathing cadence, with the most precise estimates observed at 12 cpm in nearly all postures. In both standing and sitting conditions, the RC and LC devices frequently outperformed the HMD, particularly at higher cadences, emphasizing the advantage of arm-mounted sensors in capturing respiratory motion. Angular velocity (*AngVel*) and pitch-related signals consistently emerged as the most informative features. While self-paced breathing resulted in moderate accuracy, guided breathing at 12 cpm consistently yielded the best performance. Post-exercise and drawing conditions introduced higher variability and error, particularly in standing and seated drawing tasks, although the HMD device maintained relatively stable accuracy. Overall, RR estimation was shaped by the interaction of breathing rate, posture, and sensor placement.

4.1 Heart Rate

Standing (pre- and post-exercise): During the self-paced respiratory rate activity, the most accurate heart rate estimates were obtained using *LinAcc_Y* from the RC device (RMSE: 15.2 bpm, PAE: 15.2%). At a guided respiratory rate (RR) of 6 cpm, *AngAcc_X* from both the HMD and LC devices provided the best accuracy (RMSE: 11.0 bpm, PAE: 12.3%). When the guided RR increased to 12 cpm, RC again performed best using *LinAcc_Z* (RMSE: 12.3 bpm, PAE: 12.7%). **Post-exercise** the HMD device achieved the highest accuracy with *AngAcc_X* (RMSE: 20.1 bpm, PAE: 20.1%).

Sitting: The RC device consistently outperformed others during the self-paced activity, yielding the lowest error with *AngVel_X*

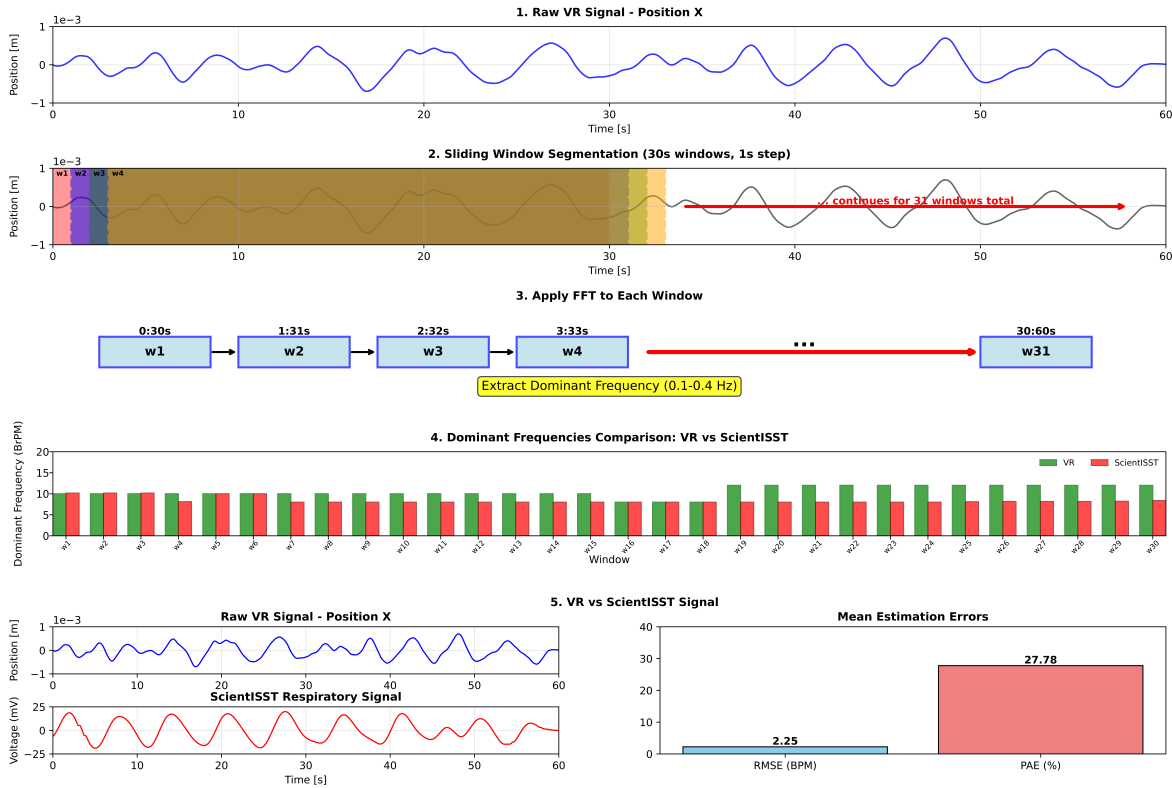


Figure 4: Signal processing pipeline used to estimate vital signs from VR device signals, illustrated using RR *Pos_X* as a case example. The same steps are applicable to HR estimation with corresponding adjustments in signal selection and filtering.

(RMSE: 7.7 bpm, PAE: 8.8%). At 6 cpm, RC maintained top performance using *LinVel_Y* (RMSE: 10.8 bpm, PAE: 12.2%). At 12 cpm, HMD performed best with *AngVel_X* (RMSE: 7.7 bpm, PAE: 8.7%).

Lying: RC achieved the highest accuracy during the self-paced activity using *LinVel_Y* (RMSE: 7.4 bpm, PAE: 12.1%). At both 6 cpm and 12 cpm guided breathing, the LC device showed the lowest error using the same feature—*LinVel_Y* (RMSE: 7.0 bpm, PAE: 9.9% and RMSE: 9.2 bpm, PAE: 14.0%, respectively).

Drawing Standing: The HMD device led across all postures. In the self-paced activity, *LinAcc_Y* performed best (RMSE = 13.0 bpm, PAE = 13.8%). At 6 cpm, *LinAcc_Z* and *LinVel_Z* were most accurate (RMSE = 13.3 bpm, PAE = 14.0%), while at 12 cpm, *AngAcc_X* yielded the best results (RMSE = 14.0 bpm, PAE = 14.7%).

Drawing Seated: During the self-paced activity, the Pen device was most accurate with *AngVel_X* (RMSE: 8.6 bpm, PAE: 10.8%). At 6 cpm, it performed best with *AngVel_Y* (RMSE: 9.8 bpm, PAE: 12.6%), while at 12 cpm, the HMD device yielded the lowest error using *AngAcc_X* (RMSE: 10.1 bpm, PAE: 11.5%).

4.2 Respiratory Rate

Standing (pre- and post-exercise): During the self-paced respiratory rate activity, the most accurate RR estimates were achieved by the RC device using *AngVel_X* (RMSE: 2.1 cpm, PAE: 16.9%).

Under guided RR at 6 cpm, the RC device delivered the most precise estimates with the *Pitch* variable (RMSE: 2.4 cpm, PAE: 23.8%). When the guided RR increased to 12 cpm, overall accuracy improved, with the lowest errors from LC's *Yaw* and RC's *Pitch* variables (both RMSE: 1.1 cpm, PAE: approximately 12.5%). **Post-Exercise** the RR estimation became more challenging, with increased error across all devices. The HMD device showed the best performance using *AngVel_Z* (RMSE: 3.6 cpm, PAE: 25.2%).

Sitting: During the self-paced activity, RC's *AngVel_Y* provided the lowest estimation error (RMSE: 2.0 cpm, PAE: 15.1%). At 6 cpm guided RR, comparable results were observed from HMD (*Yaw*), LC (*PosZ*), and RC (*Roll*), all with RMSEs around 2.8 cpm and PAEs near 30%. At 12 cpm, HMD, LC, and RC showed consistent results, each with RMSE: 0.9 cpm and PAEs between 9.3% and 9.5%.

Lying: The LC device yielded optimal performance during the self-paced activity using *AngVel_Y* (RMSE: 2.3 cpm, PAE: 17.5%). At 6 cpm, LC showed the lowest error with the same feature (RMSE: 2.3 cpm, PAE: 17.5%). At 12 cpm, LC again achieved the lowest error in Lying with *AngVel_Y* (RMSE: 1.0 cpm, PAE: 14.0%), with RC's *Pitch* variable showing similar accuracy.

Drawing Standing: In the self-paced activity, HMD with *AngVel_Z* was most accurate (RMSE = 3.4 cpm, PAE = 27.0%). At 6 cpm, HMD with *LinAcc_Z* and *LinVel_Z* performed best (RMSE = 2.7 cpm, PAE

Table 2: Best-performing kinematic features for HR and RR estimation along with their corresponding error metrics: Root Mean Square Error (RMSE) in beats per minute (bpm) and Percentage Absolute Error (PAE) in percent (%), across different VR devices (Head Mounted Display (HMD); Left Controller (LC); Right Controller (RC); Pen), activities (Self-Paced; Guided Breathing at cadence levels 6 and 12 cycles per minute) and body positions (Standing, Sitting, Lying, Drawing Standing, and Drawing Seated). The kinematic features are labeled as Position (Pos), Roll angle (Roll), Pitch angle (Pitch), Yaw angle (Yaw), Linear Velocity (LinVel), Angular Velocity (AngVel), Linear Acceleration (LinAcc), and Angular Acceleration (AngAcc). Non-acquired data are indicated by dashes. Supplementary Materials provide the full range of RMSE and PAE for all 72 signal types and VR activities.

Heart Rate																
Breathing Cadence	Device	Standing			Sitting			Lying			Drawing Standing			Drawing Seated		
		Var	RMSE	PAE	Var	RMSE	PAE	Var	RMSE	PAE	Var	RMSE	PAE	Var	RMSE	PAE
Self-Paced	HMD	LinAcc_Y	15.8	15.7	AngVel_X	8.2	9.2	AngVel_X	8.6	10.1	LinAcc_Y	13.0	13.8	LinVel_Z	10.1	11.2
	LC	LinAcc_X	15.9	15.4	LinVel_Y	8.0	8.7	AngVel_X	12.5	16.2	-	-	-	-	-	-
	RC	AngAcc_X	15.2	15.2	AngVel_X	7.7	8.8	LinVel_Y	7.4	12.1	-	-	-	-	-	-
	Pen	-	-	-	-	-	-	-	-	-	LinVel_Y	19.2	19.7	AngVel_X	8.6	10.8
Guided Breathing (6 cpm)	HMD	AngAcc_X	11.0	12.3	LinAcc_Z	11.5	13.4	Roll	13.9	16.4	LinAcc_Y	13.3	14.0	LinVel_Z	11.5	13.4
	LC	LinAcc_Z	11.0	12.3	LinAcc_Y	11.8	14.3	LinVel_Y	7.0	9.9	-	-	-	-	-	-
	RC	AngAcc_X	11.6	13.2	LinVel_Y	10.8	12.2	LinVel_X	16.4	18.5	-	-	-	-	-	-
	Pen	-	-	-	-	-	-	-	-	-	AngVel_Z	21.2	24.3	AngVel_Y	9.8	12.6
Guided Breathing (12 cpm)	HMD	AngAcc_Y	13.9	14.4	AngVel_X	7.7	8.7	AngVel_X	13.9	15.9	AngAcc_X	14.0	14.7	AngAcc_X	10.1	11.5
	LC	LinAcc_Z	14.0	14.4	AngVel_X	8.7	10.2	LinVel_Y	9.2	14.0	-	-	-	-	-	-
	RC	LinAcc_Z	12.3	12.7	LinVel_Y	10.7	12.0	AngVel_Z	14.5	17.1	-	-	-	-	-	-
	Pen	-	-	-	-	-	-	-	-	-	AngAcc_X	21.7	23.3	AngVel_X	13.4	15.9
Post-Exercise	HMD	AngAcc_X	20.1	20.1	-	-	-	-	-	-	-	-	-	-	-	-
	LC	AngAcc_X	21.0	21.3	-	-	-	-	-	-	-	-	-	-	-	-
	RC	AngAcc_X	21.3	21.4	-	-	-	-	-	-	-	-	-	-	-	-
Respiratory Rate																
Breathing Cadence	Device	Standing			Sitting			Lying			Drawing Standing			Drawing Seated		
		Var	RMSE	PAE	Var	RMSE	PAE	Var	RMSE	PAE	Var	RMSE	PAE	Var	RMSE	PAE
Self-Paced	HMD	LinVel_Z	2.8	22.8	Pos_Y	2.3	16.4	AngVel_X	3.0	24.1	LinVel_Z	3.4	27.0	AngVel_Z	2.9	25.1
	LC	AngVel_X	2.5	20.6	AngVel_Y	2.1	16.8	AngVel_Y	2.3	17.5	-	-	-	-	-	-
	RC	AngVel_X	2.1	16.9	AngVel_Y	2.0	15.1	AngVel_Y	2.5	18.9	-	-	-	-	-	-
	Pen	-	-	-	-	-	-	-	-	-	LinVel_Y	3.6	25.8	LinVel_Y	3.6	26.8
Guided Breathing (6 cpm)	HMD	Pos_Z	2.8	34.6	Yaw	2.8	31.5	Roll	2.9	32.3	PosZ	2.7	33.4	Pos_Z	2.6	31.6
	LC	Roll	3.0	31.0	Pos_Z	2.8	31.2	LinVel_Z	2.8	42.4	-	-	-	-	-	-
	RC	Pitch	2.4	23.8	Roll	2.8	30.4	LinVel_X	3.4	49.7	-	-	-	-	-	-
	Pen	-	-	-	-	-	-	-	-	-	Roll	4.2	51.9	AngVel_Y	3.3	41.9
Guided Breathing (12 cpm)	HMD	Pos_Z	1.3	13.8	PosX	0.9	9.5	Pitch	1.4	14.2	Pos_Z	2.0	20.3	Pos_X	1.1	11.4
	LC	Yaw	1.1	12.4	Yaw	0.9	9.3	AngVel_Y	1.0	11.4	-	-	-	-	-	-
	RC	Pitch	1.1	12.8	Yaw	0.9	9.5	Pitch	1.0	10.2	-	-	-	-	-	-
	Pen	-	-	-	-	-	-	-	-	-	Pos_Y	1.5	15.5	AngVel_Z	1.6	13.4
Post-Exercise	HMD	LinAcc_X	3.7	27.1	-	-	-	-	-	-	-	-	-	-	-	-
	LC	AngVel_Z	3.6	25.2	-	-	-	-	-	-	-	-	-	-	-	-
	RC	AngVel_Z	4.1	28.5	-	-	-	-	-	-	-	-	-	-	-	-

= 33.4%), while at 12 cpm, *HMD* with *AngAcc_X* yielded the highest accuracy (RMSE = 1.5 cpm, PAE = 15.5%).

Drawing Seated: For the self-paced activity, *HMD*'s *AngVel_Z* produced the best results (RMSE: 2.9 cpm, PAE: 25.1%). At 6 cpm, *HMD* again performed best with *PosZ* (RMSE: 2.6 cpm, PAE: 31.6%).

At 12 cpm, *HMD*'s *PosX* was most effective (RMSE: 1.1 cpm, PAE: 11.4%).

4.3 Use Case: the *Ghostethoscope*

Given the promising results for estimating HR and RR across the various VR activities (Table 2), we explored a use case to demonstrate

Ghost's practical applicability by converting a VR controller into a stethoscope for physiological monitoring in VR: the *Ghoststethoscope*. To this end, a volunteer (male, 46 years old) was instrumented with ECG sensors and a respiratory band, and equipped with a Meta Quest 3 HMD together with the right controller, which was used as a digital stethoscope while standing and sitting (Figure ??), as previously described in **Section 3.5**, but in this use case we also considered apnea. The table contained in Figure ?? lists the results. In each posture, the volunteer held the controller close to the chest to simulate cardiorespiratory monitoring. The same virtual environment with the animated ring (**Section 3.5**) was used for self-paced and guided breathing. Upon receiving an audio cue from a team member, data collection began and continued for 2 minutes. The volunteer was first instructed to breathe at three different cadences: self-paced, 6 cpm, and 12 cpm. A 30 s rest was provided between breathing conditions, and a 2-minute rest was given between postures. For the apnea condition, the same standing and seated postures were repeated. However, this time, the participant held their breath for 20–30 s, after verbal instructions.

Standing (without and with apnea): The most accurate HR estimates during the self-paced respiratory activity were obtained using the *LinVel_Y* variable (RMSE of 11.0 bpm and a PAE of 12.0%). Under guided breathing at 6 cpm, *AngVel_X* yielded the highest precision (RMSE: 0.8 bpm, PAE: 1.0%), while Roll produced the lowest error at the 12 cpm rate (RMSE: 8.5 bpm, PAE: 9.4%). During **apnea**, the *LinAcc_Y* showed the best performance with a RMSE of 10.6 bpm and a PAE of 11.0%. For RR estimation, *Pos_Y* demonstrated the best performance during the self-paced condition (RMSE: 0.9 cpm, PAE: 10.6%). In the 6 cpm guided breathing task, *Pos_X* delivered the most accurate estimates (RMSE: 0.4 bpm, PAE: 7.0%), followed closely by *Pos_Y* at 12 cpm (RMSE: 0.2 bpm, PAE: 1.4%).

Sitting (without and with apnea): Heart rate estimates during the self-paced respiratory activity were most accurate when using the *Roll* and *Yaw* variables, each resulting in an RMSE of 0.984 bpm and a PAE of 1.1%. At a guided breathing cadence of 6 cpm, *Roll* remained the top-performing feature (RMSE: 3.4 bpm, PAE: 4.9%), whereas at 12 cpm, *Pos_Z* offered the greatest accuracy (RMSE: 2.2 bpm, PAE: 2.7%). Under the **apnea** condition, the *Pos_Y* variable delivered the most accurate respiratory rate estimates, achieving an RMSE of 0.4 cpm and a PAE of 0.5%.

5 Discussion

Our study demonstrates the feasibility of *Ghost* to estimate HR and RR solely from the kinematic data of consumer-grade, standalone VR devices. These estimates were derived across various user postures and breathing cadences, using measurements of position, orientation, linear and angular velocity, and acceleration from each device and validated against a gold-standard method. Although validation could have been performed with PPG wristbands, we used ECG to ensure highly accurate and reliable reference data. This choice was motivated by evidence that wristband PPG signals can be less accurate and by limited access to raw PPG data; ECG provided greater low-level control over data collection and analysis. We provide a detailed mapping between these features and their associated estimation errors (Table 2). In contrast to complex

signal processing or machine learning models, we proposed *Ghost*, a simple yet robust signal processing approach based on FFTs to extract periodic signatures corresponding to HR and RR. Our signal processing framework is purposefully simple and can be extended to support real-time deployment in future work.

Overall, the most effective features for HR estimation were predominantly angular metrics, especially *AngVel_X* and *AngAcc_X*, which is consistent with previous findings by Floris et al. [10] and Solbiati et al. [28]. In contrast, RR estimation exhibited an inverse trend, where position and orientation features ranked highest in accuracy, though several velocity-based features also showed strong performance. Interestingly, while position and orientation features were not the top performers for HR estimation, they frequently ranked within the top 5 features across many activities (see Supplementary Materials). Therefore, these results underscore the physiological relevance embedded in subtle, involuntary micro-movements captured by modern VR motion tracking systems to extract vital signs related to heart and respiration movements during common VR activities.

The results also reveal several distinct patterns that can serve as practical design guidelines for physiological monitoring across common VR activities, which include standing (pre- and post-exercise), sitting, lying, and drawing with self-paced or guided-breathing, each showing unique signal and device performance characteristics as detailed below.

Standing and Lying: Estimation accuracy was strongly dependent on both posture and breathing cadence, corroborating observations by Solbiati et al. [28] that sitting and supine positions reduce motion artifacts and improve signal quality. In our data, the sitting posture consistently produced the lowest RMSE and PAE values across all activities, particularly for the RC device during self-paced RR (RMSE: 7.7 bpm, PAE: 8.8%) and the HMD device at 12 cpm (RMSE: 7.7 bpm, PAE: 8.7%). These results are comparable to those reported by Bach et al. [2] using contactless HR monitoring methods.

Lying and Drawing Standing: Accelerometer-derived variables such as *LinAcc_Y* and *LinVel_Y* occasionally outperformed, particularly in dynamic postures like lying or in Drawing Standing postures using RC and LC devices. This suggests that signal relevance is influenced by motion context and anatomical attachment. Amir et al. [1] similarly emphasized the importance of context-specific signal selection in wearable systems.

Drawing Standing and Drawing Seated: The strong performance of the *Pen* device highlights the potential of leveraging localized micro-movements for HR estimation, supporting the concept introduced by Amir et al. [1], who embedded real-time signal processing into smart eyewear to exploit subtle bodily motion.

Self-Paced and Guided Breathing: Guided respiratory activities at 6 and 12 cpm generally improved HR estimation compared to non-guided activities. This improvement may stem from reduced physiological variability and more regular rhythms during paced respiration, which enhances signal periodicity and supports more robust spectral feature extraction. Carpi et al. [6] similarly achieved less than 4% RR error using synchronized audio-visual cues.



		Heart Rate					
Breathing Cadence	Device	Var	Standing		Sitting		
			RMSE	PAE	Var	RMSE	PAE
Self-Paced	RC	LinVel_Y	11.0	12.0	LinAcc_Z	0.5	0.6
Guided Breathing (6 cpm)	RC	AngVel_X	0.8	1.0	Pos_Y	0.2	0.2
Guided Breathing (12 cpm)	RC	Roll	8.5	9.4	Yaw	4.6	5.6
Apnea	RC	LinAcc_X	10.6	11.0	Pos_Y	0.4	0.5

		Respiratory Rate					
Breathing Cadence	Device	Var	Standing		Sitting		
			RMSE	PAE	Var	RMSE	PAE
Self-Paced	RC	Pos_Y	0.9	10.6	Pos_Z	0.8	6.4
Guided Breathing (6 cpm)	RC	Pos_X	0.4	7.0	LinVel_X	0.5	4.4
Guided Breathing (12 cpm)	RC	Pos_Y	0.2	1.4	Roll	0.0	0.0

Figure 5: The *Ghostethoscope* setup, with a volunteer placing a controller on his chest for HR and RR monitoring. The adjacent table summarizes the best-performing kinematic features extracted from the Right Controller (RC) for estimating HR and RR across standing and sitting postures under different breathing cadences, including apnea (listed values from just a single volunteer). Reported error metrics include Root Mean Square Error (RMSE) and Percentage Absolute Error (PAE).

Self-Paced and Guided Breathing: In the Post Exercise condition, the HMD’s *AngAcc_X* showed a marked decline in accuracy (RMSE: 20.1 bpm). This deterioration likely reflects motion-induced artifacts, a known limitation of both video-based [32, 35] and inertial methods. Techniques such as Deep-Filter [32] or adaptive signal separation algorithms may help mitigate motion noise under high-activity. In addition, performance declined notably post-exercise, with HR errors approaching 20 bpm and RR errors exceeding 3.7 cpm (PAE: 27.1%). This degradation mirrors challenges noted by Amir et al. [1], who found high physiological variability and movement intensity to undermine signal quality. Compared to high-end gyroscopes or video-based rPPG methods, which can reach HR MAEs near 2 bpm [2], consumer VR offer a trade-off between accuracy and the convenience of a self-contained, external camera-free system compatible with various user scenarios.

Left and Right Controllers: Our findings indicate that controllers, particularly the right controller, produced HR RMSEs as low as 7.7 bpm (e.g., sitting; HMD) and RR RMSEs below 10 cpm (e.g., RC *LinVel_Y*, RMSE: 10.7 cpm at 12 cpm RRs), suggesting that controllers can supplement headset data [29].

Taken together, these observations and patterns underscore the capacity of consumer-grade VR systems to extract vital sign metrics across a range of activities, though performance may be limited in high-motion scenarios. Compared to high-end gyroscopes or video-based rPPG systems—capable of reaching HR MAEs near 2 bpm [2], VR offers a practical trade-off between accuracy and the convenience of a self-contained, camera-free platform suitable for diverse user contexts.

This study has several limitations that we must acknowledge. First, the relatively small, homogeneous sample of healthy young adults limits generalizability, excluding variability in older adults or those with cardiovascular, respiratory, or mobility conditions. Moreover, information on handedness was collected but not analyzed, and we recognize this as an additional limitation. However, this homogeneity enabled a controlled evaluation, positioning the study as one of the few investigations of its kind. Second, although the structured activities used reflect common VR scenarios, the protocol comprised only short, restful, predefined activities and may not capture the complexities of long-term or highly interactive VR experiences that require large ranges of motion (e.g., VR fitness). Third, the study focused exclusively on a single VR platform (i.e., Meta Quest 3 devices), limiting the applicability of findings across the broader landscape of consumer VR hardware. Fourthly, we did not explore signal fusion approaches nor did we apply advanced signal processing techniques like machine learning-based denoising, adaptive filtering, sensor fusion or multi-axis analysis. These methods could significantly enhance estimation accuracy and robustness, particularly in high-motion conditions, especially during postures involving large ranges of movement.

6 Conclusions

This study demonstrates that HR and RR can be reliably estimated using only the subtle micro-movements captured by consumer-grade VR headsets, controllers, and pens. By leveraging kinematic data, our *Ghost* approach enables non-contact physiological monitoring within immersive environments across popular VR scenarios,

including standing, seated, and supine postures, as well as post-exercise recovery. Notably, HR estimation was most accurate when using angular features, while RR estimation was more effectively captured through position and orientation metrics. These findings highlight the presence of physiologically meaningful signals embedded in involuntary movements fluctuations. By eliminating the need for external biosensors, our approach simplifies setup and preserves user comfort, hence, transforming standalone VR devices into biosensing platforms capable of detecting core vital signs.

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